CS 584 Natural Language Processing

Schaefer School of Engineering & Science/Dept. of Computer Science

Instructor: Justo Karell
Course Schedule: Monday-Sunday
Contact Info: jkarell@stevens.edu
Virtual session URL: Use the Zoom link in the course menu to access live session info
Prerequisite(s): Undergraduate: MA 222 or MA 232; graduate: permission of instructor

This course is fast-paced and covers a lot of ground, so it is important that you have a solid foundation on both the theoretical and empirical fronts. You should have a background in python programming, probability theory, linear algebra, calculus, and foundations of machine learning.

COURSE DESCRIPTION

Natural language processing (NLP) is one of the most important technologies referring to automatic computational processing of human languages. This includes algorithms that take human-produced text as input or produce text as output. People communicate almost everything in language: emails, phone calls, language translation, web searches, reports, books, social media, etc. Human language is symbolic in nature and also highly ambiguous and variable. Comprehending human language is a crucial and challenging part of artificial intelligence. There are a large variety of underlying tasks and machine learning models behind NLP applications. Recently, deep learning approaches have been studied and achieved high performance in many NLP tasks. The course provides an introduction to machine learning and deep learning research applied to NLP. We will cover topics including word vector representations, neural networks, recurrent neural networks, convolutional neural networks, seq2seq models, as well as some attention-based models.

STUDENT LEARNING OUTCOMES

After successful completion of this course, learners will be able to:

1. Implement gradient descent (GD) and stochastic gradient descent (SGD) techniques for learning problems
2. Apply word2vec models in real-world text corpora for business applications
3. Implement neural network models and backpropagation optimization in platforms such as TensorFlow
4. Apply dependency parsing, recurrent neural networks, and convolutional neural networks in NLP
5. Apply sequence to sequence models and attention in NLP deep neural networks

COURSE FORMAT AND STRUCTURE

This course is fully online. To access the course, please visit LUMINA. For more information about course access or support, contact the Beacon Staff.

Course Logistics

- You are encouraged to "mentally enroll" in this course as if it occurred on Mondays [Or whatever day of the week you have chosen]. In other words, our weeks will run from Monday to Sunday. I will post information (online activities, discussion starters, etc.) for the upcoming week by Sunday evening so that when you log in on Monday, you can begin the new week.
- When assignments are due, they are due by Sunday at 11:59pm on the due date listed in the course schedule.
- Deadlines are an unavoidable part of being a professional, and this course is no exception. Course requirements must be completed and posted or submitted on or before the specified due date and delivery time deadline. Due dates and delivery time deadlines are in Eastern Time (as used in Hoboken, NJ). Please note that students living in distant time zones or overseas must comply with this course time and due date deadline policy. Avoid any inclination to procrastinate. To encourage you to stay on schedule, due dates have been established for each assignment; 20%of the total points will be deducted for assignments received at most 2 days late; assignments received more than 2 days late will receive 0 points.

Instructor’s Online Hours

Virtual Office Hours will be held via weekly synchronous Zoom sessions to discuss questions related to weekly readings and/or assignments.

TENTATIVE COURSE SCHEDULE

Module	Topic(s)	Readings	Assignment
Module 01	Introduction to Natural Language Processing	None	Python Knowledge Check (quiz)
Module 02	Machine Learning Basics	None	Python Coding #1 Due
Module 03	NLP Language Representation	Required: SLP Ch. 2	Begin Project 1
Module 04	Language Models and Sentiment Classification	Required: SLP Ch. 3 and 4	Python Coding #2 Due
Module 05	Regression and Vector Semantics	Required: SLP Ch. 5 and 6	Project 1 Due
Module 06	Neural Networks and Sequence Labeling	Required: SLP Ch. 7 and 8	Begin Project 2 Python Coding #3 Due
Module 07	Deep Learning Architectures	Required: SLP Ch. 9	Work on Project 2
Module 08	Machine Translation	Required: SLP Ch. 10	Project 2 Due
Module 09	Transfer Learning and Constituency Grammars	Required: SLP Ch. 11 and 12	Begin Project 3
Module 10	Constituency and Dependency Parsing	Required: SLP Ch. 13 and 14	Work on Project 3
Module 11	Question Answering and Visualization Paths	Required: SLP Ch. 23	Project 3 Due
Module 12	Chatbots and Dialogue Systems	Required: SLP Ch. 24	Begin Project 4
Module 13	Automatic Speech Recognition	Required: SLP Ch. 26	Project 4 Due

COURSE MATERIALS

Textbook(s): Speech and Language Processing, 3rd edition draft, by Jurafsky and Martin, 2022

Materials: Python 3.5 or greater with the Jupyter IDE

COURSE REQUIREMENTS

As you move through the topics in each module, rather than simply viewing lecture content, you will be asked to read, listen to, and watch a variety of media. You'll also be regularly prompted to actively evaluate your knowledge as you're building it.

The components in each module are designed to be completed sequentially in order. In addition to videos, readings and interactives, here are a few types of learning activities you'll encounter.

- Knowledge Checks: These activities evaluate your mastery of concepts shortly after they are introduced. You are expected to complete these checks. These activities help you prepare for assignments.
- Discussion Board: In every module, you'll find a discussion board in which you are encouraged to pose and answer questions with regard to the concepts presented in the module. We may also discuss these questions in the live sessions.

TECHNOLOGY REQUIREMENTS

Baseline technical skills necessary for online courses

- Basic computer and web-browsing skills
- Navigating Canvas

Technology skills necessary for this specific course

- Live web conferencing using Zoom
- Python programming

Required Equipment

- Computer: current Mac (OS X) or PC (Windows 7+) with high-speed internet connection
- Webcam: built-in or external webcam, fully installed
- Microphone: built-in laptop or tablet mic or external microphone

Required Software

- Microsoft Word
- Python 3.5 or greater with the Jupyter IDE

GRADING PROCEDURES

Grades will be based on:

Homework	40%
Projects (4 @ 15% each)	60%
TOTAL	100%

Course Grade:

Letter Grade	Scale
A	100% to 94%
A-	< 94% to 90%
B+	< 90% to 87%
B	< 87% to 84%
B-	< 84% to 80%
C+	< 80% to 77%
C	< 77% to 70%
F	< 70% to 0%

Late Policy

20% of the total points will be deducted for assignments received 1 - 2 days late; assignments received more than 2 days late will receive 0 points.

Academic Integrity

Graduate Student Code of Academic Integrity

All Stevens graduate students promise to be fully truthful and avoid dishonesty, fraud, misrepresentation, and deceit of any type in relation to their academic work. A student's submission of work for academic credit indicates that the work is the student's own. All outside assistance must be acknowledged. Any student who violates this code or who knowingly assists another student in violating this code shall be subject to discipline.

All graduate students are bound to the Graduate Student Code of Academic Integrity by enrollment in graduate coursework at Stevens. It is the responsibility of each graduate student to understand and adhere to the Graduate Student Code of Academic Integrity. More information including types of violations, the process for handling perceived violations, and types of sanctions can be found at www.stevens.edu/provost/graduate-academics.

Special Provisions for Undergraduate Students in 500-level Courses

The general provisions of the Stevens Honor System do not apply fully to graduate courses, 500 level or otherwise. Any student who wishes to report an undergraduate for a violation in a 500-level course shall submit the report to the Honor Board following the protocol for undergraduate courses, and an investigation will be conducted following the same process for an appeal on false accusation described in Section 8.04 of the Bylaws of the Honor System. Any student who wishes to report a graduate student may submit the report to the Dean of Graduate Academics or to the Honor Board, who will refer the report to the Dean. The Honor Board Chairman will give the Dean of Graduate Academics weekly updates on the progress of any casework relating to 500-level courses. For more information about the scope, penalties, and procedures pertaining to undergraduate students in 500-level courses, see Section 9 of the Bylaws of the Honor System document, located on the Honor Board website.

Undergraduate Honor System

Enrollment into the undergraduate class of Stevens Institute of Technology signifies a student's commitment to the Honor System. Accordingly, the provisions of the Stevens Honor System apply to all undergraduate students in coursework and Honor Board proceedings. It is the responsibility of each student to become acquainted with and to uphold the ideals set forth in the Honor System Constitution. More information about the Honor System including the constitution, bylaws, investigative procedures, and the penalty matrix can be found online at http://web.stevens.edu/honor/

The following pledge shall be written in full and signed by every student on all submitted work (including, but not limited to, homework, projects, lab reports, code, quizzes and exams) that is assigned by the course instructor. No work shall be graded unless the pledge is written in full and signed.

"I pledge my honor that I have abided by the Stevens Honor System."

<u>Reporting Honor System Violations</u>

Students who believe a violation of the Honor System has been committed should report it within ten business days of the suspected violation. Students have the option to remain anonymous and can report violations online at www.stevens.edu/honor.

LEARNING ACCOMMODATIONS

Stevens Institute of Technology is dedicated to providing appropriate accommodations to students with documented disabilities. The Office of Disability Services (ODS) works with undergraduate and graduate students with learning disabilities, attention deficit-hyperactivity disorders, physical disabilities, sensory impairments, psychiatric disorders, and other such disabilities in order to help students achieve their academic and personal potential. They facilitate equal access to the educational programs and opportunities offered at Stevens and coordinate reasonable accommodations for eligible students. These services are designed to encourage independence and self-advocacy with support from the ODS staff. The ODS staff will facilitate the provision of accommodations on a case-by-case basis.

For more information about Disability Services and the process to receive accommodations, visit https://www.stevens.edu/office-disability-services. If you have any questions please contact: Phillip Gehman, the Director of Disability Services Coordinator at Stevens Institute of Technology at pgehman@stevens.edu or by phone 201-216-3748.

Disability Services Confidentiality Policy

Student Disability Files are kept separate from academic files and are stored in a secure location within the Office of Disability Services. The Family Educational Rights Privacy Act (FERPA, 20 U.S.C. 1232g; 34CFR, Part 99) regulates disclosure of disability documentation and records maintained by Stevens Disability Services. According to this act, prior written consent by the student is required before our Disability Services office may release disability documentation or records to anyone. An exception is made in unusual circumstances, such as the case of health and safety emergencies.

INCLUSIVITY

Name and Pronoun Usage

As this course includes group work and class discussion, it is vitally important for us to create an educational environment of inclusion and mutual respect. This includes the ability for all students to have their chosen gender pronoun(s) and chosen name affirmed. If the class roster does not align with your name and/or pronouns, please inform the instructor of the necessary changes.

Inclusion Statement

Stevens Institute of Technology believes that diversity and inclusiveness are essential to excellence in academic discourse and innovation. In this class, the perspective of people of all races, ethnicities, gender expressions and gender identities, religions, sexual orientations, disabilities, socioeconomic backgrounds, and nationalities will be respected and viewed as a resource and benefit throughout the semester. Suggestions to further diversify class materials and assignments are encouraged. If any course meetings conflict with your religious events, please do not hesitate to reach out to your instructor to make alternative arrangements.

You are expected to treat your instructor and all other participants in the course with courtesy and respect. Disrespectful conduct and harassing statements will not be tolerated and may result in disciplinary actions.

MENTAL HEALTH RESOURCES

Part of being successful in the classroom involves a focus on your whole self, including your mental health. While you are at Stevens, there are many resources to promote and support mental health. The Office of Counseling and Psychological Services (CAPS) offers free and confidential services to all enrolled students who are struggling to cope with personal issues (e.g., difficulty adjusting to college or trouble managing stress) or psychological difficulties (e.g., anxiety and depression). Appointments are can be made by phone (201-216-5177).

EMERGENCY INFORMATION

In the event of an urgent or emergent concern about the safety of yourself or someone else in the Stevens community, please immediately call the Stevens Campus Police at 201-216-5105 or on their emergency line at 201-216-3911. These phone lines are staffed 24/7, year round. For students who do not reside near the campus and require emergency support, please contact your local emergency response providers at 911 or via your local police precinct.

Other 24/7 national resources for students dealing with mental health crises include the National Suicide Prevention Lifeline (1-800-273-8255) and the Crisis Text Line (text “Home” to 741-741). If you are concerned about the wellbeing of another Stevens student, and the matter is *not* urgent or time sensitive, please email the CARE Team at care@stevens.edu. A member of the CARE Team will respond to your concern as soon as possible.

.